

Pre-registered report: Mapping Synesthetic Sequences in Space: Improving Consistency Test by Harnessing Cartography Tools.

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Sequence-space synesthesia is a perceptual condition in which ordinal sequences (e.g., numbers, days of the week, or months of the year) are experienced as occupying specific spatial locations. This perceptual condition is identified using self-reports (e.g. questionnaires) and behavioural consistency tests. Existing consistency tests measure the spatial distance between responses to repeated stimuli, but they ignore the ordinal and geometric features of spatial-forms. This pre-registered study consists of a present and future analysis. In the present analyses, we optimise SSS diagnostics from consistency test by extracting novel geometric features at the spatial-form level. To do this, we first evaluate classification performances of old and new features in a large ($N = 685$) aggregated sample from four available and independent datasets. We then harness a geographic toolbox to quantify topological spatial-forms. Finally, receiver operating characteristic (ROC) analyses are conducted to assess the discriminative performances of each method. The results show that permuted topological validity of spatial-forms and the perimeter between stimuli performs equally well in detecting SSS. In a future study, we will examine predictive validity of the selected methods on an independent dataset that has yet to be collected. Findings from the present study highlight the relevance of topological principles for sequence-space representations and aims to optimise the ability of consistency tests to detect SSS.

Keywords: Sequence space synaesthesia, Synaesthesia/synesthesia, consistency test, space, time, numbers

Introduction

Sequence-Space Synaesthesia (SSS) is a subtype of synaesthesia where visuo-spatial-forms is the phenomenon where people visualize ordered sequences in particular spatial positions. For example, numbers, weekdays or months (synesthetic *inducers*) are represented as arranged into specific spatial positions in space (synesthetic *concurrent*). The synaesthetic spatial-forms are characteristic and form complex and variable geometric forms. Those geometric forms might be different for each categories (i.e. number-forms

weekdays and months), with forms such as ellipses, zig-zags or curved lines Flournoy (1893). SSS are usually identified using questionnaires and consistency tests. An individual is identified as SSS based on the variability between spatial across repeated responses to the same inducers: less variable responses being an indicator of the consistent response characteristic of SSS. However, stimulus level consistency can also be achieved by responding to the same position for all stimuli. Furthermore, it might favour linear over circular spatial-forms and is not informed about the ordinality between stimuli (e.g. $1 \rightarrow 2 \rightarrow 3$, etc.). Here, we aim at optimizing the identification of SSS from consistency tests using geometrical characteristics of spatial-forms of ordered stimuli or inducers.

SSS frequently co-occurs with other subtypes of synaesthesia and as for other subtypes of synaesthesia it's idiosyncratic. Colour-grapheme synaesthesia where a grapheme *inducer* triggers a *concurrent* colour (i.e. "A" seen as red), co-occurs at 71 % to 76 % with SSS (Sagiv et al., 2006). Nevertheless, across different synaesthetic subtypes, self-reported SSS tend to cluster together (Ward & Simner, 2022), indicating a degree of internal homogeneity and justifying to study it as a distinct phenomenon. Spatial-forms are idiosyncratic, which results in considerable heterogeneity in how this phenomenon is manifested across individuals. One source of heterogeneity is given by dimensionality: some SSS experi-

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ences spatial-forms involve three-dimensional (3D) and two-dimensional (2D) arrangements (Eagleman, 2009; Price & Pearson, 2013). Another source is the reference frame, for example, the spatial-forms might take place in an external space around the body (*i.e.* projector) or in an egocentric internal space (*i.e.* associator) (Dixon et al., 2004; Smilek et al., 2007). Further variability can be explained by temporal-spatial properties and mental-manipulation of spatial-forms such as “zooming” in and out or rotating or shifting perspectives at will (Gould et al., 2014; Jarick et al., 2009). Lastly, the shape, complexity and layout of the spatial-forms are also heterogeneous such as forming for example ovals, lines, zig-zags or loops. With some recurring shapes being more frequent across SSS, such as ovals for months (Eagleman, 2009).

Despite the after mentioned heterogeneities, SSS is also phenomenological characterized, there might be five main characteristics for synaesthesia in general and specifically SSS (Deroy & Spence, 2013; Seron et al., 1992). *Automaticity*: the inducer automatically triggers the concurrent. *Unidirectionality*: while the *inducer* triggers the concurrent, the concurrent does not trigger the inducer. *Developmentally*: the experience was already present during childhood. *Consciousness*: The concurrent is consciously perceived. *Consistency*: the inducer-concurrent pair remains stable in time within subject. While some of these characteristics can be captured with self-reported questionnaire (*i.e.* for development and consciousness), other can be quantified more objectively using behavioural tests such as consistency tests (Baron-Cohen et al., 1993).

Consistency tests

Consistency tests are behavioural measures used to evaluate how stable synesthetic experiences over time (*e.g.*, Rothen et al., 2013; Rothen et al., 2016). They do this by presenting the same stimuli repeatedly and assessing how similar a participant’s responses are across repetitions (*i.e.* *inducer-concurrent* associations). High consistency is often taken as evidence for synaesthesia. Consistency tests have proven effective for colour-grapheme synaesthesia. Measures of individual consistency can be derived using colour pickers while repeatedly presenting the same inducer (*i.e.* “A”). The quantification of the distance between concurrents from the same inducers is used to discriminate consistent synaesthetes from controls. Specifically, the euclidean distance in CIE L*u*v colour space, which is designed for perceptual uniformity, yields optimal classification accuracy when using a cut-off estimated as deduced from a larger representative sample (Rothen et al., 2013). A similar rationale to that used for colour-grapheme synaesthesia has been applied to characterize SSS from consistency tests.

In the SSS consistency test, participants complete multiple rounds of the same stimuli (*e.g.*, digits, days of the week, and months of the year) and are asked to indicate a spatial lo-

cation for each stimulus by clicking on the computer screen. Brang et al. (2010) for example evaluated consistency as the distance between repeated responses to the same inducer or stimuli (*e.g.* January and January) relative to adjacent stimuli (*e.g.* February). A response was defined as consistent if it fell within 1.96 *z-scores*. However, this criterion was noted to be potentially too conservatory, as it identified synaesthesia in 4 of 81 self-reported synaesthetes. Geometrical features such as area and perimeter between the coordinates of repeated inducers have further been used as a measure of the distance between same inducers, hence as a metric for consistency. Less consistent responses leading to smaller triangle perimeters and area (Rothen et al., 2016a). A cut-off corresponding to less than 0.203 % average responses area in proportion to the total screen area to classify as SSS has been suggested (see also Ward et al., 2018).

However, several general caveats of consistency tasks using concurrent or stimulus level metrics have been identified. Some geometrical features may favour certain spatial-forms (Ward et al., 2018) in particular those with linear or highly regular spatial layouts, *i.e.* elliptical patterns for months (Brang et al., 2010; Eagleman, 2009; Flourenoy, 1893). Another concern is that participants that respond with the same position, for example when not knowing the response, will have artificially high consistencies with those metrics (Rothen et al., 2016a). This has led to alternative approaches such as to add the standard deviation of responses, questionnaire cut-off (Ward et al., 2018) or permutation-based comparison of individual responses to chance levels for colour-grapheme (Root et al., 2021) and SSS (Ward, 2022). Finally, consistency methods evaluate the variability of responses to the same stimuli, therefore the ordinality between stimuli is not taken into account in these measures. Synesthetic spatial-forms follow ordinal rules (*e.g.* Monday → Tuesday → Wednesday). In the domain of numerical cognition, ordinality has been identified as an important information when processing sequences (Lyons & Beilock, 2013). Some studies of SSS have systematically investigated ordinality by investigating metrics of adjacent inducer-concurrent pairs such as distances (Brang et al., 2010) or angles (Eagleman, 2009). However, ordinality of the concurrent spatial-form might be particularly relevant considering all the stimuli of a certain category (*i.e.* weekdays or months). The spatial-forms of the concurrents are often spatially structured into configuration such as lines or polygons that may follow distinctive geometrical rules. For example linear layouts have been described in early accounts of number forms (Flourenoy, 1893; Galton, 1880).

Present study

The goal of this registered report is to compare different consistency features on their classification performances of individuals with SSS and controls using Receiver Opera-

From the merged datasets, we kept 685 from the total 689 participants. First, we excluded 0.02 % empty trials (i.e. skipped responses) including trials flagged for having the same x or y coordinates across conditions and repetitions, causing the depletion $n = 2$ participants (as in Rothen et al., 2016b; Ward et al., 2018). $n = 2$ participants were excluded for having less than 4 coordinate points since polygons need at least four coordinate. Note therefore, that 31 participants of the final sample did not have responses in all three conditions X repetition cases. From the final sample of $N = 685$, 396 were synaesthetes and 289 controls, see Table 1 and Table 2.

Table 1*Synaesthetes*

Synaesthetes	N before	N after	Age
(Rothen et al., 2016a)	33	32	23.1
(Van Petersen et al., 2020)	23	22	23.2
(Ward, 2022)	252	252	37.2
Ward 2	90	90	NA
Total:	398	396	

Note. Synaesthetes. Demographics were not provided in the dataset Ward 2. Before: before participant exclusion

Table 2*Controls*

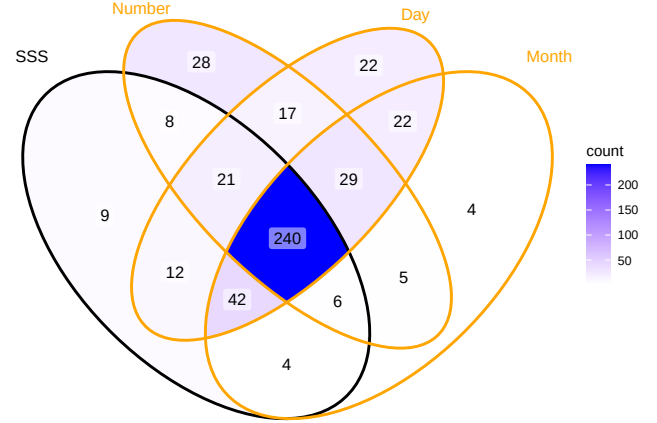
Controls	N before	N after	Age
(Rothen et al., 2016a)	37	37	28.2
(Van Petersen et al., 2020)	21	21	21.6
(Ward, 2022)	215	213	19.9
Ward 2	18	18	NA
Total	291	289	

Note. Controls. Demographics were not provided in the dataset Ward 2. Before: before participant exclusion

Regarding the synaesthesia profiles, detailed participant information such as questionnaire responses were only available from the two datasets from the data by Professor Ward (i.e. a majority of 573 participants). We described the self-reported profiles for the stimulus categories used in the consistency test (i.e. number, weekdays and month), see Figure 2. From this Venn-diagram we see that 240 of the SSS report having spatial-forms for Numbers, Days and Month; 42 only Days and Month; 21 only Number and Day. Also, 127 controls report spatial-forms for either Days, Numbers, Months or combinations.

Figure 2

Venn diagram of the types of self-reported SSS



Note. Only data from Pr. Ward is represented here. SSS = Sequence-space synaesthesia

Procedure and analysis

For the consistency test, each stimuli were presented randomly and sequentially centrally on the screen. Participant were instructed to click on the screen position where they visualize them. In Van Petersen et al. (2020) and Rothen et al. (2016a) the participants were allowed to skip responses.

Stimulus included here were 7 weekdays (Monday to Sunday), 12 months (January to December) and 9 numbers (0 to 9). For all the data from Professor Ward the stimuli were presented in randomized order with the constraint that no stimulus was repeated until all unique stimuli ($N = 29$) had been presented once. The median display resolution was 1440 X 768, with a maximum of 2560 X 2025 and a minimum of 308 X 149.

Code and data for “one-click” reproducibility of the following analyses are available on git-hub (<https://github.com/remLach/SpaceSequenceSynDiagnostic>).

Stimulus level: area and perimeter between repetitions

First, we reproduced features extracted from consistency tests found in the literature such as area and perimeter (Rothen et al., 2016a; Van Petersen et al., 2020; Ward, 2022). Formally, the area (see Equation 1) and perimeter (see Equation 2) are calculated from the triangle formed by the x and y coordinates of the three repetitions of each stimuli (i.e. $(xI,$

$y1), (x2, y2), (x3, y3))$.

$$Area = \frac{(x1y2 + x2y3 + x3y1 - x1y3 - x2y1 - x3y2)}{2} \quad (1)$$

$$Perimeter = \sqrt{(x2 - x1)^2 + (y2 - y1)^2} + \sqrt{(x3 - x2)^2 + (y3 - y2)^2} + \sqrt{(x1 - x3)^2 + (y1 - y3)^2} \quad (2)$$

Each stimuli's area was then averaged by participants. The area metric is in % of the screen area to be able to compare with the consistency across studies using different screen sizes. To account for screen size and spread variability across the datasets and participants, area and perimeter were also calculated on individually normalized x and y coordinates (*z-score*). These methods compute consistency metrics at the stimulus level, the rationale is that consistent responses would show less variability between repetitions. For example, smaller area or perimeter of the triangle formed indicate more variability hence more consistent responses between the three repetitions.

Form level: novel features

First, we extracted features at the spatial-form level for each repetition separately. Taking the stimuli as an ordered sequence we can consider them as a geometrical segments (i.e. open geometrical form) and polygons (i.e. closed geometrical form), similarly as originally described in Galton (1880). The first extracted feature is self-intersection. We considered the spatial-forms as a segment constituted by the lines connecting consecutive stimuli separately for each category and repetitions (i.e. Monday (repetition 1) → Tuesday (repetition 1) → Wednesday (repetition 1),) and then counted the number of times each segment self-intersect. The rationale is that SSS should have less chance to produce that self-intersect than control. The number of self-intersections were computed separately for each repetitions and conditions and averaged per participants.

Second, we harnessed a geospatial analysis, the 'sf' package version 1.0.21 (Pebesma & Bivand, 2023) to extract geometry-based features from participant's 2D (x,y) coordinate responses. This package allows, for example, to build and analyse segments or polygons and then extract multiple geometrical descriptors or features. Informed by the ordinality of the stimulus, we defined segments and polygon by conditions and repetitions. The rationale here was to determine whether, when considering the spatial-forms formed by stimuli as ordered coordinates (i.e. as segments or polygon) they remain consistent *between* repetitions for each individuals. We extracted the topological validity, topological simplicity and a compactness score for each polygons.

Topological validity

We assessed topological validity using geometric validation test. Topological validity tests if a polygons is well-formed and valid according to the Open Geospatial Consortium (OGC) Simple Features Specification (Herring, 2010). A polygon is considered topologically valid if it satisfies the following criteria: (1) if polygon rings are simple (i.e., they do not touch or self-intersect), (2) boundary rings to not cross (3) boundary rings may only touch points tangentially (4) rings that define holes are contained within the exterior ring (5) the polygon rings must not splits the polygon (PostGIS 3.6.2dev Manual, n.d.).

For each participant we tested for topological validity across individual 3 categories (weekdays, month and numbers) and 3 repetitions, hence 9 forms. The binary outcome (1 = is valid, 0 = invalid) was then averaged across all 9 forms. Hence, it can span from 1 (all 9 are valid) to 0 (none of the 9 forms are valid). Thus, the topological validity score ranges from 0 (none of the nine forms are valid) to 1 (all nine forms are valid). For example, if a participant's forms for weekdays were valid across all three repetitions, months were valid in two of three repetitions, and numbers were invalid in all repetitions, the topological validity score would be $5/9 \approx 0.56$.

Topological simplicity

We assessed topological simplicity, which evaluates whether a geometric object has a simple structure without self-intersections or self-tangencies. According to the OGC Simple Features Specification (Herring, 2010), a polygon boundary is considered simple if it does not pass through the same point more than once (PostGIS 3.6.2dev Manual, n.d.). For polygons, simplicity requires that each ring does not self-intersect or self-touch. Note that simplicity is a condition for validity: a polygon can be simple but still invalid (e.g., if an interior ring extends outside the exterior ring).

Compactness score

Finally, we also implemented a measure of compactness for the space-forms. We used the score developed to quantify the gerrymandering of political district by Polsby and Popper (1991). Gerrymandering is a politically motivated strategy of redefining district's boundary to give the advantage to a political party. This has the consequences to reduce the compactness of those districts as would be expected if they where defined from geomorphology. The compactness is calculated as the ratio of the area and perimeter of a polygon. The rationale is that more compact forms (i.e. higher Polsby–Popper scores) would be more likely in SSS than controls. Formally, the Polsby–Popper score is calculated from the polygon's area and perimeter Equation 3:

$$score = \frac{4\pi \cdot Area}{Perimeter^2} \quad (3)$$

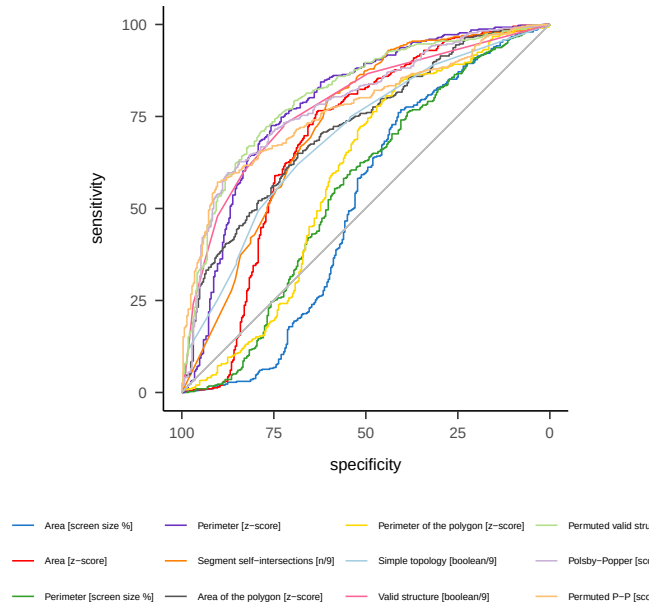
$$D = \frac{AUC_1 - AUC_2}{SD} \quad (5)$$

We compared all ROC's to the one with the highest AUC.

Results

Figure 3

Receiver Operating Characteristic (ROC) curves of all features



Note. Grey line indicates chance level

Descriptively, the triangle area (in % screen size) between repetition for SSS was surprisingly larger than in previous studies: 0.26% ($SD = 0.52\%$) compared to 0.15% ($SD = 0.13\%$) in Rothen et al. (2016a) and 0.14 % ($SD = 0.17\%$) in Ward et al. (2018), see Table 4. These values also differ between the available datasets (See Figure B2). These differences could be mitigated by using individually standardized coordinates $M = 0.05$ ($SD = 0.06$) Rothen et al. (2016a), $M = 0.08$ ($SD = 0.15$) Ward (2022), $M = 0.09$ ($SD = 0.15$), Ward2, see also Table 4.

ROC analyses

All results of the ROC analyses are summarized in Table 3 and Figure 3. All ROC are compared to the permutated validity method, since it led the highest AUC (81.16), all p-values false discovery rate (FDR) adjusted. Interestingly, while the permutated validity leads to significantly larger AUC than the Polsby-Popper method, ($D = 2.08$, $p = 0.04$), it led to similar AUC than the standardized perimeter ($D = 1.41$, $p < .05$).

Table 3

ROC analyses of all features

	Feature	AUC	DP	threshold	sensitivity	specificity
10	Permutated valid structure [boolean/9]	81.16	2.13	0.17	70.78	78.55
11	Polsby-Popper [score/9]	78.89	2.29	0.19	59.70	87.54
4	Perimeter [z-score]	78.80	2.06	1.73	72.54	75.78
9	Valid structure [boolean/9]	78.12	1.90	0.14	72.80	72.32
12	Permutated P-P [score/9]	77.15	2.46	0.20	57.18	90.31
5	Segment self-intersections [n/9]	72.93	1.75	1.17	79.85	60.21
6	Area of the polygon [z-score]	72.03	1.36	1.29	64.99	68.51
8	Simple topology [boolean/9]	69.74	1.24	0.28	61.96	68.51
2	Area [z-score]	69.27	1.68	0.08	76.32	63.32
7	Perimeter of the polygon [z-score]	59.12	1.29	10.46	83.88	41.87
3	Perimeter [screen size %]	55.34	0.68	0.03	76.07	38.75
1	Area [screen size %]	50.87	0.79	0.21	76.83	40.48

Furthermore, the permutated validity led to significantly higher AUC than the non permutated one ($D = 3.06$, $p < .05$). Hence the two best AUC performances are from the averaged permutated validity score (AUC = 81.16; DP = 2.13, cut-off = 0.17 corresponding to 1.53 valid of 9 forms) and the other for standardized perimeter between the repetitions of each stimuli (AUC = 78.8; DP = 2.06, cut-off = 1.73 z-scores). However, at the proposed cut-off, the permutated validity leads to higher specificity than perimeter (78.55 vs. 75.78) indicating better rejection of controls (i.e. less false negatives) but lower sensitivity (70.78 vs. 72.54), indicating poorer detection of SSS (i.e. more false negatives). See Figure A1 for the densities of each standardized features suggesting bi-modal distribution.

Some of the participants might have been originally mis-attributed to control or SSS, see Figure F1. Descriptively, we also found about ~6 % of the full sample's synesthetes to be consistently classified as controls (i.e. consistent false positive) by the five methods with the highest AUC Figure F2. ~3 % controls are consistently classified as SSS (i.e. consistent

Table 4*Descriptives of each features*

Feature	Control	SSS	Control [zs]	SSS [zs]
Permuted valid structure [boolean/9]	0.12 (0.16)	0.36 (0.25)	-0.57 (0.64)	0.41 (1.01)
Polsby–Popper [score/9]	0.11 (0.13)	0.27 (0.19)	-0.51 (0.69)	0.37 (1.03)
Perimeter [z-score]	3.30 (1.72)	1.59 (1.16)	0.60 (1.04)	-0.44 (0.70)
Valid structure [boolean/9]	0.13 (0.19)	0.39 (0.28)	-0.54 (0.69)	0.39 (1.01)
Permuted P-P [score/9]	0.11 (0.17)	0.41 (0.33)	-0.56 (0.56)	0.40 (1.06)
Segment self-intersections [n/9]	9.56 (11.31)	1.87 (5.42)	0.48 (1.23)	-0.35 (0.59)
Area of the polygon [z-score]	1.00 (0.99)	1.91 (1.31)	-0.42 (0.79)	0.30 (1.03)
Simple topology [boolean/9]	0.22 (0.23)	0.40 (0.27)	-0.39 (0.85)	0.28 (1.01)
Area [z-score]	0.26 (0.32)	0.08 (0.14)	0.42 (1.29)	-0.30 (0.55)
Perimeter of the polygon [z-score]	9.49 (3.44)	8.39 (2.48)	0.21 (1.16)	-0.16 (0.83)
Perimeter [screen size %]	0.03 (0.03)	0.03 (0.08)	0.03 (0.44)	-0.02 (1.26)
Area [screen size %]	0.46 (0.87)	0.26 (0.52)	0.17 (1.25)	-0.12 (0.75)

false negative) Figure F3.

In sum, the permuted validity and perimeter both provide similarly best AUC compared to the other features. While the permuted validity is best to have fewer false positives (i.e. controls) while perimeter is best for fewer false negatives (i.e. SSS).

Further analyses

We conducted further analyses by subsampling them based on the dataset source, questionnaire percentiles. The aim of these analyses was to test the stability of the classifications from different methods across experiments (i.e. datasets) and sample characteristics (i.e. questionnaire scores). We also used a correlational approach to test for correlations between features and questionnaires.

We sub-sampled the data based on the source of the data and found different features best fit different datasets, see Figure B1. Interestingly the best features by datasets differ be-

tween the data sources. While the standardized area indeed lead to the best AUC for the data from Rothen et al. (2016a), the perimeter leads to better AUC in Ward et al. (2018). The permuted validity leads to AUC > 80 across all datasets.

To test the stability of AUC of all features, we computed them across several percentiles based on the questionnaire scores. Again, only the data from Professor Ward is included there. We computed AUC, sensitivity and sensibility on sub samples based on the percentiles of the questionnaire scores (see Ward et al., 2018), taking the 10-0% highest questionnaire scores against the 10 % worst (i.e. 90-100 %) . We further sub-sampled by 10 % questionnaire scores and recalculated AUC, sensitivity and sensibility, see Figure C1 for AUC, DP, sensitivity and specificity. Using this method we can see that the total AUC remains relatively stable across percentiles.

Another important point addressed in the literature about consistency test is circularity. The circularity is given in that if consistency tests are used to classify SSS and controls, then those groups will by definition differ in consistency (Root et al., 2025). Hence, we made a correlation matrix with the questionnaire scores and all the features, see Appendix. The two highest correlations are indeed between the questionnaire score and perimeter ($r = .58$) and permuted validity ($r = .50$), see Figure D1.

We also wanted to visualize the average space form of each category at the population level to see if there were qualitative differences between the forms. The figure in the appendix (Figure E1), shows more circular patterns for months and more linear horizontal and vertical for numbers.

Phase II Methods

Additional data will be collected in the future using the same consistency test, with the procedural exception that there will be four repetitions per stimuli instead of three (Sachdeva et al. (2024), Stage 1 in-principle acceptance). The same feature will be extracted from this dataset. Hence for the stimulus levels feature we will compare the area and perimeter of a rectangle of four coordinates pairs instead of a triangle. Materials are more details on this study are pre-registered on OSF , including ample size and recruitment method (Sachdeva et al., 2024, Stage 1 in-principle acceptance) (<https://osf.io/6wn4m>).

We will compute the same ROC analyses as in this pre-registration on all the features. Since we obtain different best AUC per features depending on the dataset (see Figure B1) it is likely that the best features here will not match with the best feature in the to be collected dataset.

Phase I. Discussion

Our investigation of four datasets found two main features leading to the optimal classification of SSS from control: standardized perimeter between repetitions and per-

mutated topological validity. While perimeter is calculated on the stimulus level, i.e. distance between repeated stimuli, topological validity is calculated on the whole spatial-form level, e.g. x,y coordinates formed by Monday to Sunday. Both criteria display bimodal distributions across the groups (see Figure A1). We also obtain variable best features by datasets, in particular from the stimulus based metrics (area and perimeter), while topological validity lead to > 80 AUC across datasets. Considering only the data by Professor Ward that contains questionnaire data, we obtain similar AUC and DP for more extreme Synaesthete and Control groups when sub-sampled by questionnaire's score distribution percentiles (see Figure C1). However, perimeter lead to slightly higher correlation with the questionnaire ($r = .57$) than permuted topological validity ($r = .51$), see Figure D1.

Limitations

Although an optimal test to classify SSS might be particularly relevant for experimental purposes, several important limitations need to be considered. First, consistency tests contain only a limited set of sequential stimuli (i.e. months, weeks and the first ten natural numbers) that could potentially elicit synaesthetic experiences. Other ordinal categories such as temperature, clock time, musical keys might be more relevant for some individuals with SSS. For numbers specifically, better consistency tests might be obtained using a larger set size (i.e. including decades and hundreds), as descriptively interesting form changes occur at different decimals in base-10 number systems (Galton, 1880).

Second, the use of diagnostic cutoffs assumes categorical distinctions between groups, but SSS may exist on a continuum and scores might be more suitable (Price & Pearson, 2013). This limitation is related to the circularity mentioned previously: diagnostic cutoffs as calculated here depend on how SSS and controls are classified in the first place (Simner, 2012). Measures of other characteristics of synaesthesia such as automaticity or visual Gabor detection might be necessary to establish external validity (Ward et al., 2018). Indeed, determining the prevalence of SSS in the general population requires definitional choices that might be based on more conservative or lenient criteria (Brang et al., 2010; Jonas & Price, 2014; Sagiv et al., 2006; Ward et al., 2018). These difficulties are reflected in the span of current prevalence estimates: 4.4 % (Brang et al., 2013), 8.1 % (Ward et al., 2018) and 14 % (Seron et al., 1992).

Third, as shown in Figure B1, we find that the optimal criteria can vary across datasets. In other words, we obtain different AUC depending on the datasets. These differences might be explained by different sampling methods biases, for example (Van Petersen et al., 2020) recruited SSS from over one hundred candidates to maximise participant reporting synaesthetic experiences. Additionally, the original classification into control and SSS are not consistent across studies and it

might be useful to have several validation measures as pointed out in the limitation (see Ward et al., 2018).

Another possible explanation may be methodological and related to the option to skip responses in the task, leaving empty cases for some participants. Indeed, having fewer responses (i.e., fewer coordinates) increases the likelihood of producing a valid spatial form also in controls. Conversely, perimeter and area are differentially affected by missing responses.

Topological validity

Surprisingly, permuted topological validity of the spatial-forms led to similar results than the perimeter (or the distance) between the responses. This result might be informative about how SSS map ordinal stimuli in space. Indeed, it seems the patterns of spatial-forms follow topological rules analogous to geographical and geometry-based space structures. Analogies between maps and neuroscience have a long history (i.e. retinotopy, sonotopy or somatotopy) (Eagleman & Goodale, 2009). Therefore spatial-forms might originate from the mapping between ordinal or sequential stimuli on idiosyncratic visuo-spatial abilities following topological rules. One of the advantages of using topological validity is that it also classifies responses with the same coordinates as invalid and hence inconsistent, while using the area and perimeter metrics they would qualify as highly consistent responses.

Intermediary Conclusion

Phase I compared traditional stimulus-level consistency measures with novel form-level geometric features for detecting SSS. We found optimal classification performance from permuted topological validity and standardized perimeter between repetitions. The success of topological validity as a diagnostic feature is theoretically meaningful: genuine synaesthetes should produce spatially coherent, well-formed structures that satisfy geometric constraints compared to controls. Similarly, the perimeter feature captures the stability of the overall spatial configuration across repetitions, which should remain consistent for true SSS individuals regardless of presentation order.

In *phase II*, we will attempt at validating the criteria on a yet-to-be acquired dataset.

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Appendix A

Feature distributions

Regarding the distributions of each features across the SSS vs. controls, Figure A1 we compared the density plots of each standardized criteria (in order to make them comparable) which visually suggests bimodal distribution.

Appendix B

ROC analysis by dataset

Here we compare the ROC analyses results for each data sample of the three methods with the best AUC, see Figure B2. Descriptively, we also looked at the distributions of each methods across SSS and controls Figure B1. Note that the dataset labelled as Ward 2 has more synaesthetes than controls (5:1 ratio).

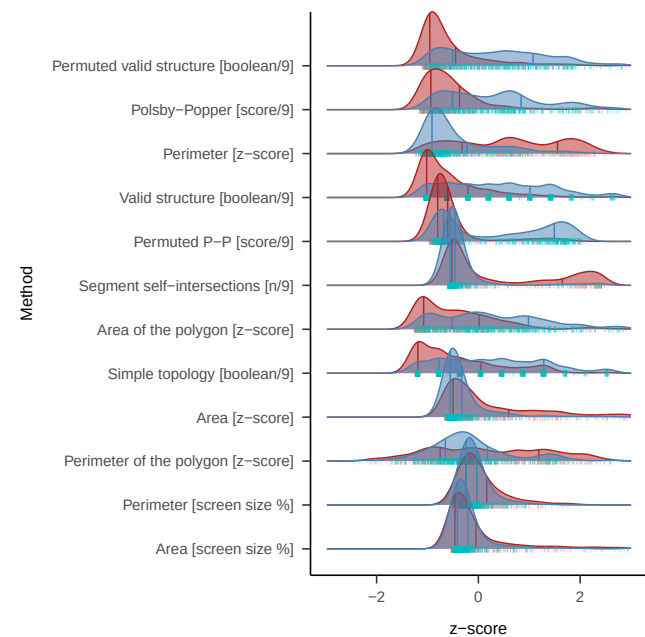
Appendix C

ROC Analysis Sub-sampled data by questionnaire quantiles (20% steps)

We compared the data sampled by the questionnaire score. Based on the distribution of the questionnaire score, we sampled the 10 % with the lowest and 10 % with the highest scores. Those are then compared with the 20 and 20 % and so on until 40 and 40 %. The rationale of this procedure is that AUC, sensitivity and specificity should remain stable across

Figure A1

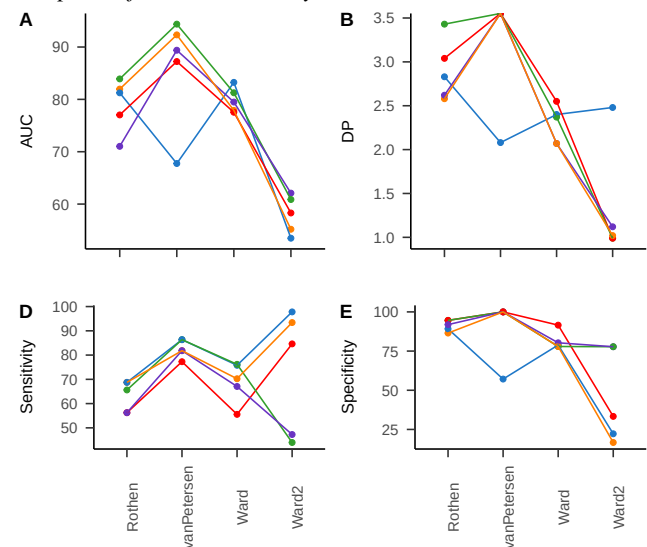
Density plots of all the features comparing SSS and controls



Note. Red = Classified as Synesthetes, Blue = as controls. All method's score have been standardized (z-score). x axis treamed between -3 and 3 z-scores. Vertical line indicate respectively 25 and 75 percentiles of each distributions.

Figure B1

Lineplots of AUC and DP by data source

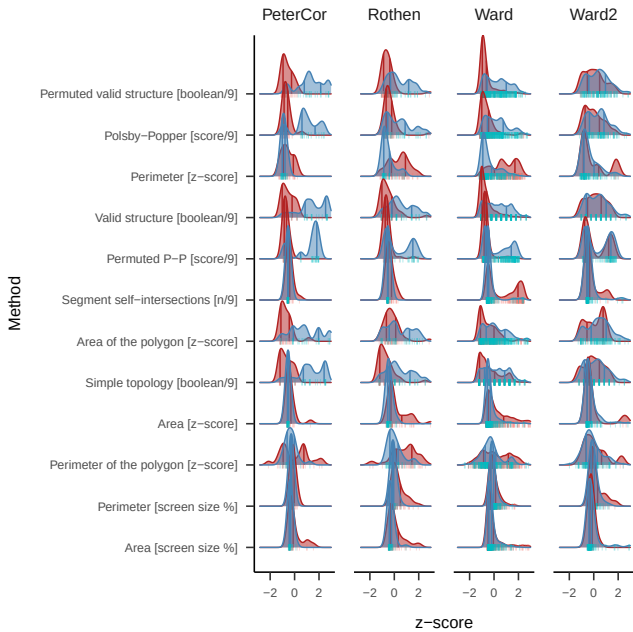


Perimeter [z-score] • Permuted P-P [score/9] • Permuted valid structure [boolean/9] • Polsby-Popper [score/9]

Note. AUC = Area Under the Curve, DP = Discrimination Power

Figure B2

Density plots of all the features comparing SSS and controls by dataset



Note. Red = Classified as Synesthetes, Blue = as controls. All method's score have been standardized (z-score). x axis treated between -3 and 3 z-scores. Vertical line indicate respectively 25 and 75 percentiles of each distributions.

percentiles for a feature to be valid, see Figure C1. In other words the ROC should remain unchanged if we take extreme groups compared to less extreme ones.

Appendix D

Correlation with self-report

The best criterion should also best correlate with SSS self-reported questionnaire score. Works only with Ward's aggregated data, see Figure D1.

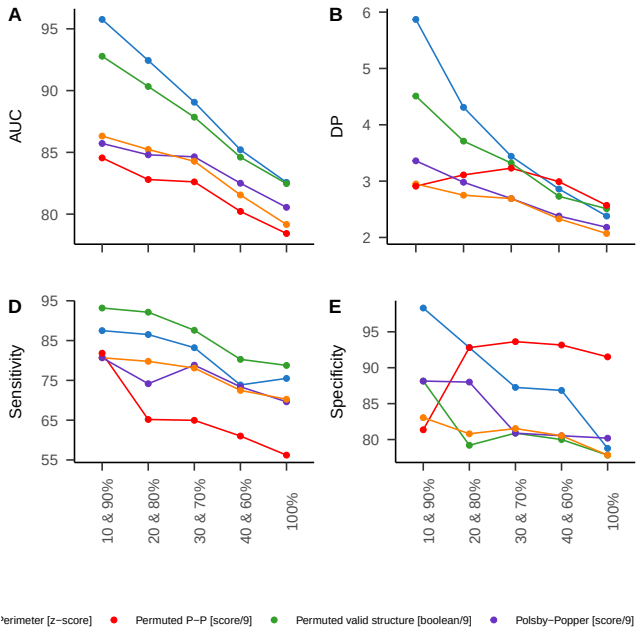
Appendix E

Centered average forms

In this additional analyses we attempted at visualizing the average for each categories to see whether a pattern could be discerned on a qualitative level. To avoid negative coordinates, we shifted all the z-scores from the whole dataset minimum for the x and y coordinates. Then we anchored to the coordinate 0,0 the stimulus "1" for numbers, "Monday" for weekdays and "January" for months. We then plotted all the space forms with transparency, see Figure E1.

Figure C1

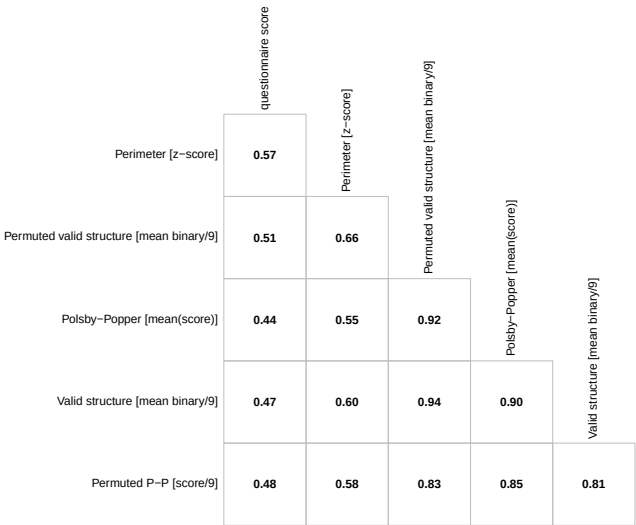
Lineplots of AUC and DP by questionnaire percentile



Note. Each point is an increasing percentiles. AUC = Area Under the Curve. DP = Discrimination Power.

Figure D1

Correlation with self-reported questionnaire

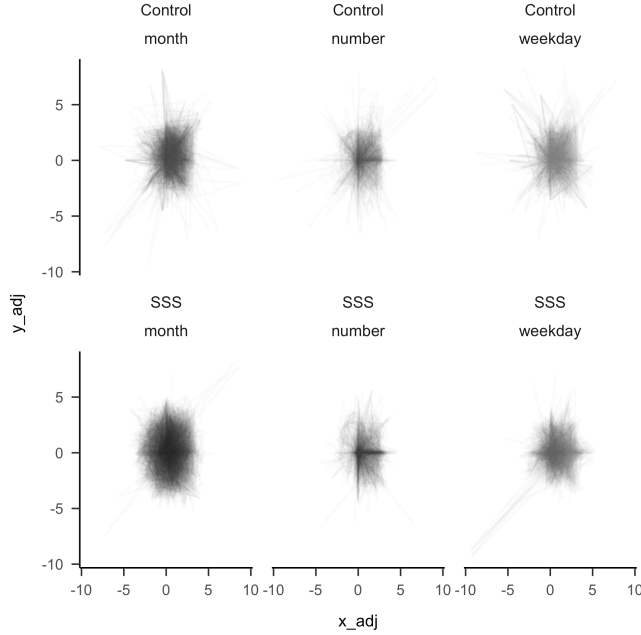


Note. Only data from Ward is included here

Descriptively it is possible to see more circular pattern for months and horizontal and vertical patterns for numbers. Also the averaged forms appear to be more clearly defined in the synaesthetes than control.

Figure E1

Average Centered space forms visualized



Appendix F

Alluvial plot

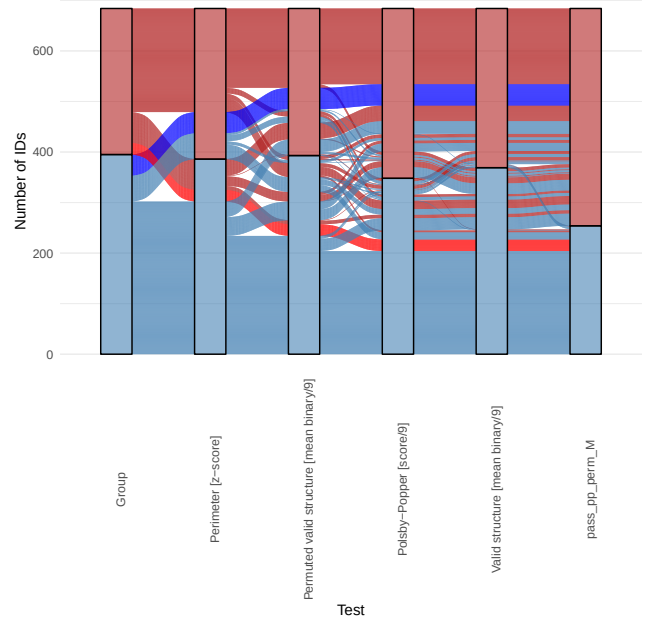
Are false positives and negatives share some characteristics? To answer this question qualitatively we identified two group of self-reported SSS participants who are consistently classified as controls by the five methods with the highest AUC (i.e. consistent false negatives), see dark blue lines in Figure F1. On the other side we identified a group of controls who are consistently classified as SSS by the five methods with the highest AUC, see dark red lines in Figure F1.

On one side 6.12 % of all participants, that is 42 Synaesthetes where consistently classified as controls by all five tests, see Figure F2. On the other side 3.35 % of all participants, 23 controls where consistently classified as SSS by all five tests, see Figure F3. Given the different approaches of each methods, the identified participants might be on one side non-genuine synaesthetes and on the other controls who have SSS but are either not aware or did not identify it as such.

Appendix G

Figure F1

Alluvial plot of test passing



Note. Red = Controls, Blue = Synaesthetes. Highlighted in blue are Synaesthetes who are classified as control by all 5 tests. Highlighted in red are controls who are classified as synaesthetes by all 5 tests.

Figure F2

Synaesthetes who consistently fail all five tests



Figure F3*Controls who consistently pass all five tests***Code to visualize all spatial-forms**

This exports many pdf's. It plots each ID and condition z-score x and y coordinates. Since each coordinate is repeated 3 times, these are represented by triangles. The line paths connect average coordinates to visualize forms (stimulus are ordered, i.e. 1 to 9, Monday to Sunday, January to December). Finally in the top right corner, each dots indicates if the ID would pass (green dot) / fails (red dot) depending on the criteria.